CS342: Computer Vision and Pattern Recognition

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Course Description

In an era where cameras and digital imagery permeate our daily lives, understanding computer vision has become essential. CS342 provides an introductory exploration into this fascinating field. From casual "selfies" to immersive augmented reality experiences, the applications of computer vision are ubiquitous.

In this course, we delve into critical questions that shape the landscape of computer vision today:How do computers efficiently search for specific images amidst vast datasets? What algorithms enable the automatic tagging of photos on social media platforms? Can machines accurately interpret natural gestures or sign languages for seamless human-computer interaction? How do self-driving cars navigate complex terrains using visual input?

Our journey spans a wide range of topics, from the basics of image processing to cutting-edge visual recognition techniques. We explore machine learning methods, including supervised algorithms and deep neural networks, to tackle intricate research problems.

Moreover, we engage in discussions about ethical considerations, privacy, and bias in the responsible development of AI-driven systems.

Prerequisite(s):

Basic knowledge of probability and linear algebra; data structures, algorithms; programming experience. Previous experience with image processing will be useful but is not assumed.

Credit Hours: 4

Textbook

The course textbook is: ***Computer Vision: Algorithms and Applications (Second Edition)*, Richard Szeliski.** It is freely available online or may be purchased in hard-copy. Course lecture slides will be posted online and are also a useful reference.
You may also find the following books useful.

- *Digital Image Processing*, Rafael Gonzalez and Richard Woods
- *Computer Vision: A Modern Approach*, David A. Forsyth and Jean Ponce
- *Computer Visio*n, Linda G. Shapiro and George C. Stockman
- *Multiple View Geometry in Computer Vision*, Richard Hartley and Andrew Zisserman.

- *Pattern classification*, Richard O. Duda, Peter E. Hart, and David G. Stork
- *Pattern Recognition and Machine Learning* Christopher M. Bishop

Course Objectives

Upon successful completion of the course, students will achieve the following learning objectives:

- Students will acquire a comprehensive understanding of the theoretical and practical aspects of computing with images, including image formation, measurement, and analysis.
- Students will implement common methods for robust image matching and alignment, demonstrating proficiency in image processing techniques.
- Students will grasp the geometric relationships between 2D images and the 3D world, enabling them to understand concepts such as image transformations, alignment, and epipolar geometry.
- Students will gain exposure to object and scene recognition and categorization from images, including techniques such as object detection, supervised classification algorithms, and probabilistic models for sequence data.
- Students will comprehend the principles of state-of-the-art deep neural networks, starting with convolutional neural networks (CNNs), and their applications in computer vision.
- Students will develop practical skills necessary to build computer vision applications, enabling them to apply learned concepts in real-world scenarios effectively.
- Students will view Computer Vision as a research area, understanding its ongoing developments, challenges, and potential applications.

By meeting these objectives, students will be equipped with the knowledge and skills needed to pursue further studies or careers in computer vision, machine learning, and related fields.

Course Outcome

- Students will demonstrate proficiency in image processing techniques, including feature extraction, filtering and edge detection through practical implementation and application exercises.
- Students will acquire the ability to employ segmentation and clustering algorithms, Hough transform, and robust fitting techniques, such as RANSAC for grouping and fitting objects and contours within images.

- Students will develop a deep understanding of multiple view geometry, local invariant feature detection and description, image transformations, planar homography, and stereo vision techniques for analyzing images and objects from different perspectives.
- Students will demonstrate competence in object and scene recognition, categorization, and classification using supervised learning algorithms, probabilistic models, support vector machines, and neural networks.
- Students will gain practical experience in building computer vision applications, utilizing learned techniques to address real-world problems, such as object detection and image classification.
- Through assignments, projects, and exams, students will develop critical thinking skills and problem-solving abilities necessary for tackling challenges in computer vision and pattern recognition domains.
- Students will develop a research-oriented perspective towards computer vision, gaining exposure to advanced topics, current trends, and emerging techniques in the field through lectures, readings, and discussions.
- Students will enhance their communication skills through project and paper presentations.
- Students will explore ethical considerations and implications related to computer vision technologies, understanding issues such as privacy, bias, and societal impact, and developing a responsible approach towards the use and development of computer vision systems.
- By the end of the course, students will recognize the importance of lifelong learning in the rapidly evolving field of computer vision, being prepared to adapt to new technologies, methods, and research findings throughout their careers.

Grade Distribution:

- **Quizzes:** 20%
- **Projects and/or Assignments:** 30%
- **Midterm Exam:** 20%
- **Final Exam:** 30%

Course Outline (tentative)

The weekly coverage might change as it depends on the progress of the class. However, you must keep up with the earlier lectures and reading assignments. There will be 10 or more quizzes throughout the semester.

Week	Topic
1	Introduction and Basics of Digital Image
2	Filtering, Edge Detection and Template Matching
3	Frequency Response and Hough Transform
4	Harris Corner, SIFT
5	Subsampling, Fitting Interpolation
6	Tracking, Binary Image
7	Camera Stereo, Multiview Geometry
8	Camera Parameters, Panorama, Disparity
9	Machine Learning in Computer Vision
10	Machine Learning to Deep Learning
11	CNN, RNN, LSTM
12	Transformers, GNN
13	Segmentation
14	Generative AI for Image Generation
15	Paper Presentations
16	Advanced topics (if time permits), Applications, Course Wrap-up